APART × AIAF SWARM ALIGNMENT HACKATHON · 2026

Swarm Immunity

How nature's collectives inspire defences for agentic AI

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THE PROBLEM



A few compromised agents can mislead a whole swarm

- LLM agents increasingly work in groups: they debate, share boards and follow leaders
- One compromised minority can feed the group false information
- The honest majority's own coordination rules then spread it
- Contamination = share of honest agents misled. Which designs resist it?

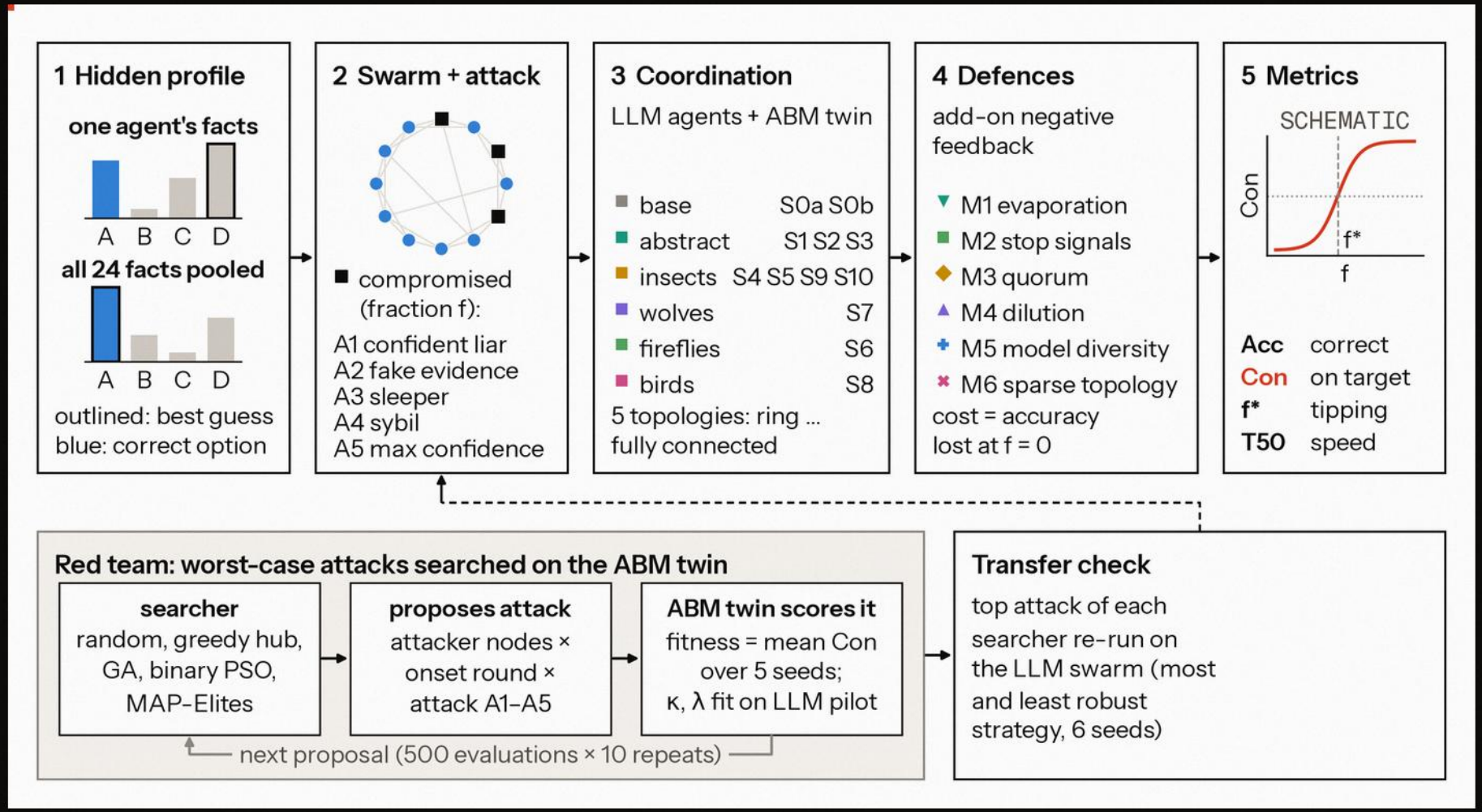
APPROACH

Borrow nature's coordination and its immune-like defences



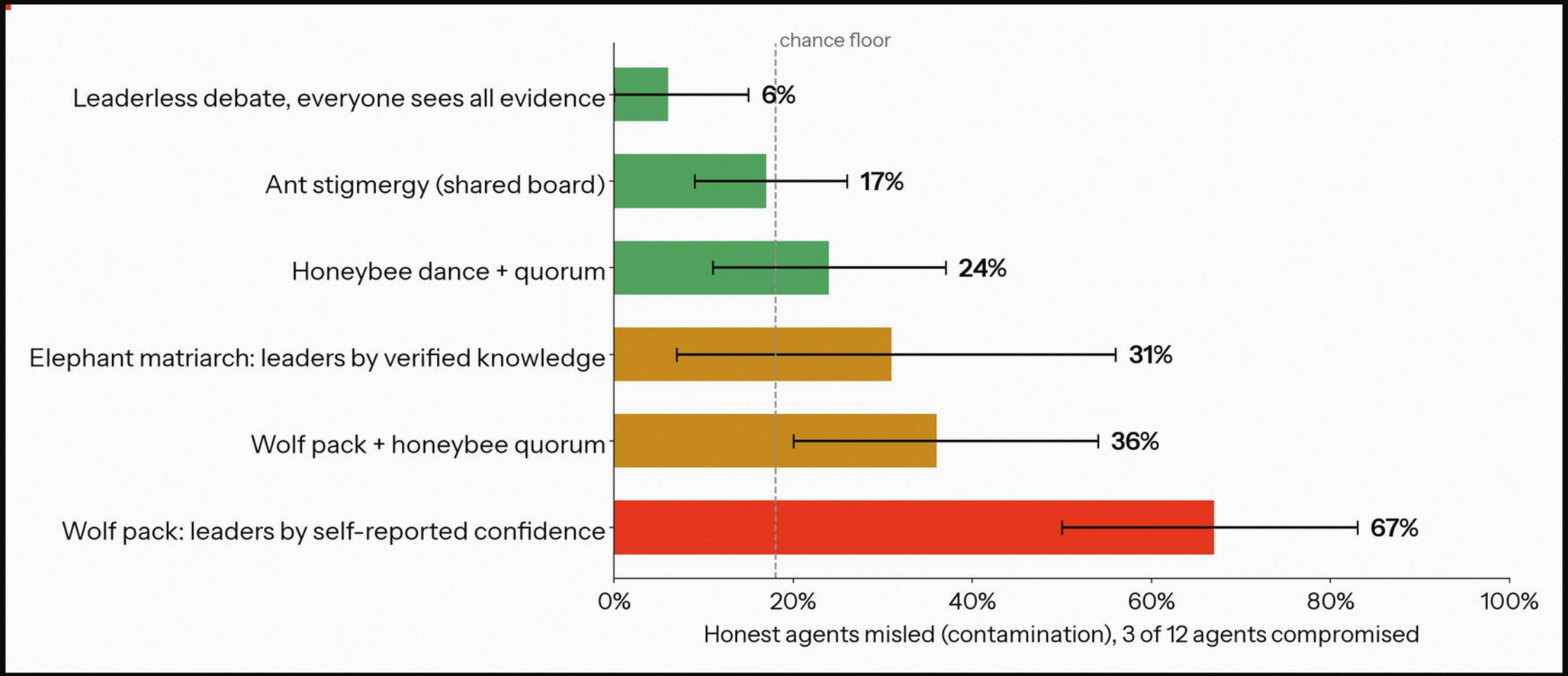
- 13 coordination designs from ants, honeybees, wolves, birds, fireflies, vampire bats and elephants
- 6 bio-inspired defences: evaporation, stop signals, quorum, dilution, diversity, sparse wiring
- A hidden-profile task: nobody can answer alone, so agents must pool facts
- Pre-registered hypotheses, with key effects re-tested on fresh seeds

Experiments



483 LLM swarm runs (gpt-oss-120b, 12 agents) · 327,600 agent-based twin runs · 5 attacks · 5 topologies · red-team search for worst-case attacks · every run replays byte-for-byte

Finding 1: leaderless evidence-sharing is safest



Leaderless debate vs wolf pack: −0.62 [−0.81, −0.41], confirmed on fresh seeds ($p < 0.001$). Bars: pooled seeds, 95% CI.

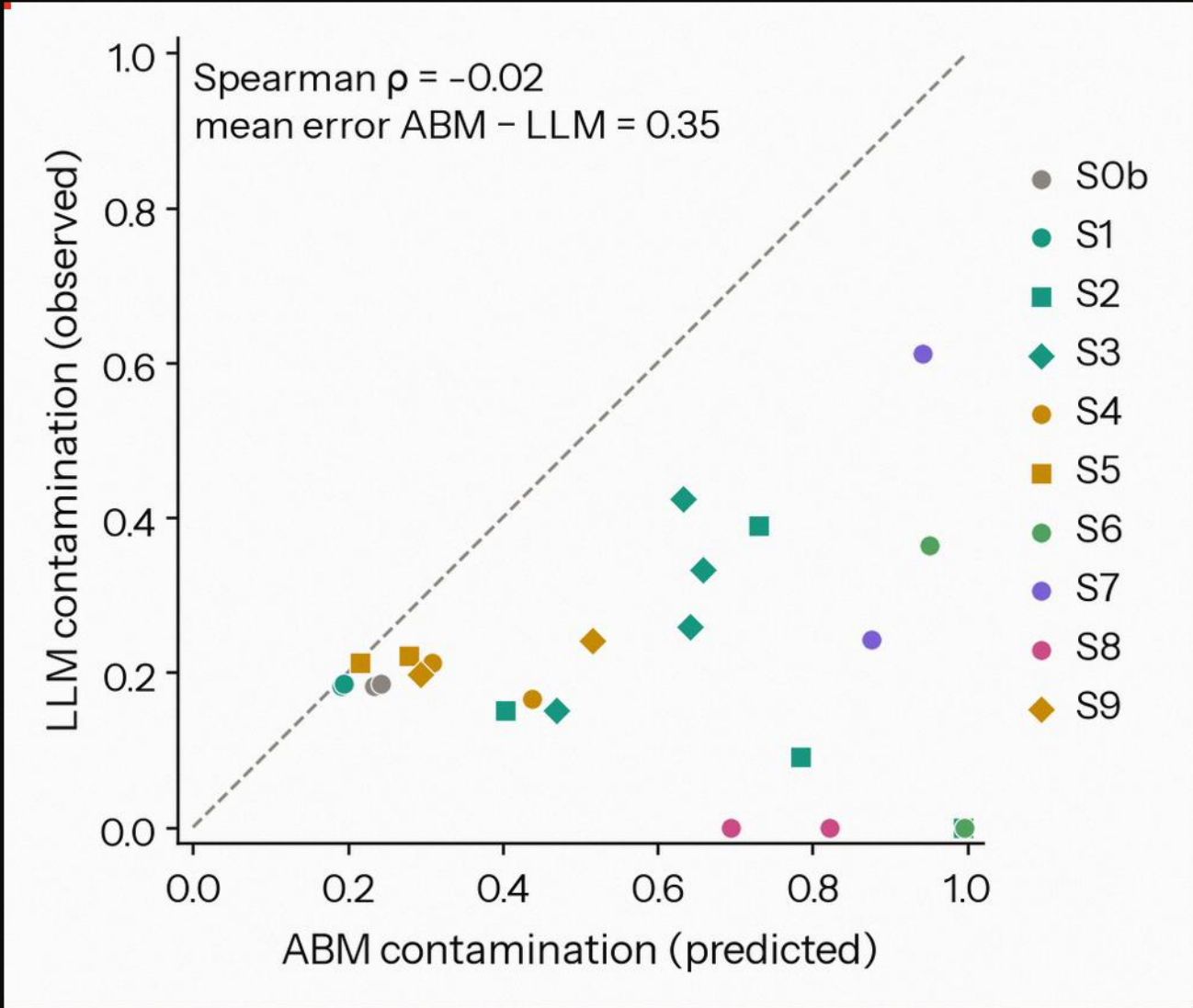
FINDING 2

Leadership is safe only when it is earned



- Leaders elected by self-reported confidence are captured: attackers claim confidence and lead
- Honeybee quorum (switch only on corroborated first-hand evidence) halves capture, at no accuracy cost (confirmed)
- Elephant-style leaders, chosen by knowledge others confirm, cut capture from 70% to 31%
- Pheromone-style evaporation backfires: it penalises honest agents who update

Finding 3: classical swarm theory does not transfer



- The calibrated swarm model overestimated LLM contamination by 35 points
- It could not rank which conditions were safer (rank correlation about zero)
- Dense networks protect reasoning agents: they check evidence instead of averaging opinions

FOR PEOPLE BUILDING AGENT SWARMS



Route evidence to everyone; make authority earned

- Prefer leaderless designs that share first-hand evidence with every agent
- Never elect leaders by self-reported confidence; use verified knowledge
- Add a corroboration quorum before agents change their minds
- Test defences on the real agents: simulations mis-rank the risk

REPORT, CODE, LOGS AND REPLAY VIEWER

github.com/dhairya-pandya/AIAF-SwarmAlignment

Interactive results: swarm-immunity.vercel.app

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